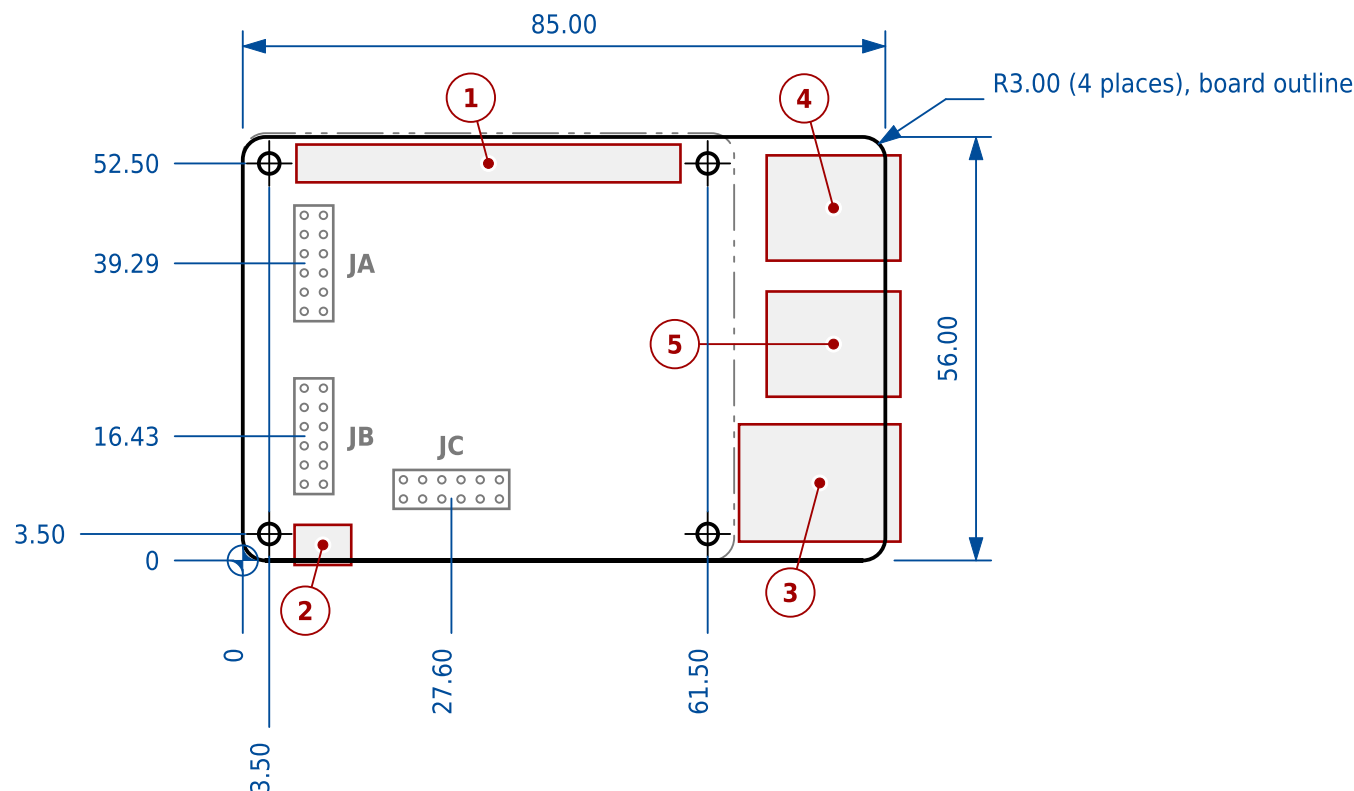




NOTES

1. Viewed from the component side.
2. Phantom outline is the Digilent Pmod HAT Adapter fitted on the 40-pin GPIO header; its mounting holes coincide with this board's.
3. Hole IDs are assigned by this drawing and mean the same hole on every Raspberry Pi sheet.
4. The Raspberry Pi 3 Model B and Raspberry Pi 3 Model B+ drawings agree on every hole and connector position, so one sheet covers both.
5. The 40-pin GPIO header position is fixed by the Raspberry Pi HAT specification, so it is the same on every Raspberry Pi sheet.
6. Design keep-outs to 6.2 mm diameter around every mounting hole, per the Raspberry Pi HAT specification; KEEPOUT gives what this model's own drawing shows.
7. Pmod host JC overhangs the lower edge, over the HDMI and power connectors. Fit the two standoffs opposite the 40-pin header, as Digilent's manual warns, so JC's pins cannot touch the HDMI shell. HDMI connectors are not drawn.

NOTES (continued)

8. Pmod HAT Adapter host positions are DERIVED, good to about +/-0.75 mm; drawing ACC-HAT-PMOD has the derivation.
9. Assembled envelope, connector overhang included: 87.00 x 56.60 mm.

SOURCES

1. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.dxf> - Raspberry Pi Ltd
2. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-plus-mechanical-drawing.dxf> - Raspberry Pi Ltd, Raspberry Pi 3 Model B+
3. Mounting holes: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf> - Drawing note on the Model B drawing, quoted: 4x M2.5 MOUNTING HOLES DRILLED TO 2.75 +/- 0.05mm. The B+ drawing gives no hole size, but its geometry is identical.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	2.75 +/-0.05	not given
MT2	61.50	3.50	2.75 +/-0.05	not given
MT3	3.50	52.50	2.75 +/-0.05	not given
MT4	61.50	52.50	2.75 +/-0.05	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	micro-USB power input	6.85 to 14.35	-0.60 to 4.71
3	Ethernet RJ45	65.65 to 87.00	2.50 to 18.00
4	USB 2.0 type A (upper pair)	69.30 to 87.00	39.65 to 53.57
5	USB 2.0 type A (lower pair)	69.30 to 87.00	21.65 to 35.57

PMOD HAT ADAPTER HOSTS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	left edge	9.400	39.290	10.670	32.940
JB	left edge	9.400	16.430	10.670	10.080
JC	bottom edge	27.600	9.400	33.950	10.670

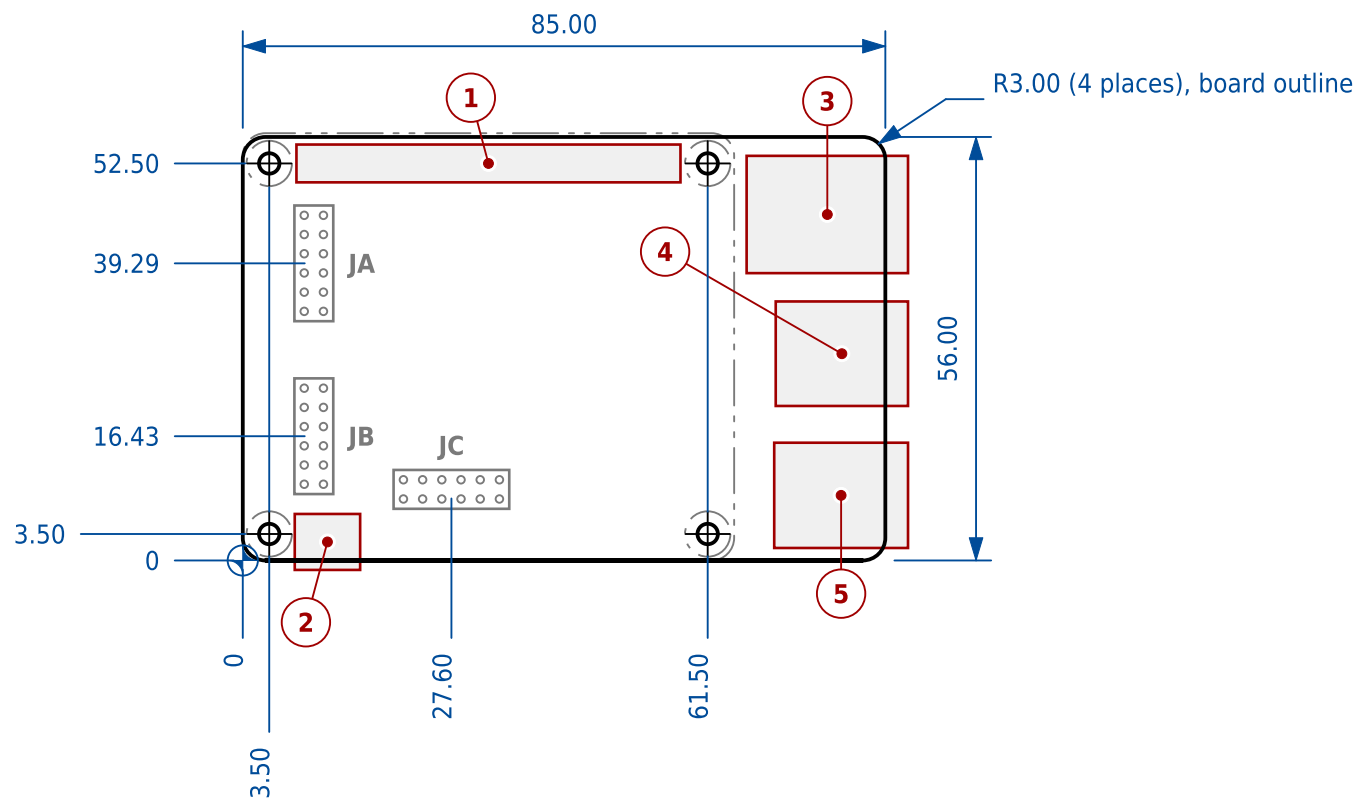
LEGEND

- Board outline
- Component body, scheduled
- Adjacent part
- Dimension, extension and leader

RASPBERRY PI 3 MODEL B AND B+

85 x 56 mm, both models

DRAWN generated	VERSION v0.0-579-g0e07c5a	UNITS mm	SCALE 1:1
SIZE A3	SHEET 1 OF 1	DRAWING NO RPI-3B	REV A
MATERIAL PCB, thickness not stated	GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia per schedule		



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	2.70	6.00
MT2	61.50	3.50	2.70	6.00
MT3	3.50	52.50	2.70	6.00
MT4	61.50	52.50	2.70	6.00

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	USB-C power input	6.88 to 15.53	-1.25 to 6.15
3	Ethernet RJ45	66.65 to 88.00	37.99 to 53.51
4	USB 3.0 type A (upper pair)	70.50 to 88.00	20.43 to 34.25
5	USB 2.0 type A (lower pair)	70.30 to 88.00	1.65 to 15.57

PMOD HAT ADAPTER HOSTS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	left edge	9.400	39.290	10.670	32.940
JB	left edge	9.400	16.430	10.670	10.080
JC	bottom edge	27.600	9.400	33.950	10.670

LEGEND

- Board outline
- Component body, scheduled
- Adjacent part, hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Phantom outline is the Digilent Pmod HAT Adapter fitted on the 40-pin GPIO header; its mounting holes coincide with this board's.
- Hole IDs are assigned by this drawing and mean the same hole on every Raspberry Pi sheet.
- The 40-pin GPIO header position is fixed by the Raspberry Pi HAT specification, so it is the same on every Raspberry Pi sheet.
- Design keep-outs to 6.2 mm diameter around every mounting hole, per the Raspberry Pi HAT specification; KEEPOUT gives what this model's own drawing shows.
- Pmod host JC overhangs the lower edge, over the HDMI and power connectors. Fit the two standoffs opposite the 40-pin header, as Digilent's manual warns, so JC's pins cannot touch the HDMI shell. HDMI connectors are not drawn.
- Pmod HAT Adapter host positions are DERIVED, good to about +/-0.75 mm; drawing ACC-HAT-PMOD has the derivation.

NOTES (continued)

- Assembled envelope, connector overhang included: 88.00 x 57.25 mm.

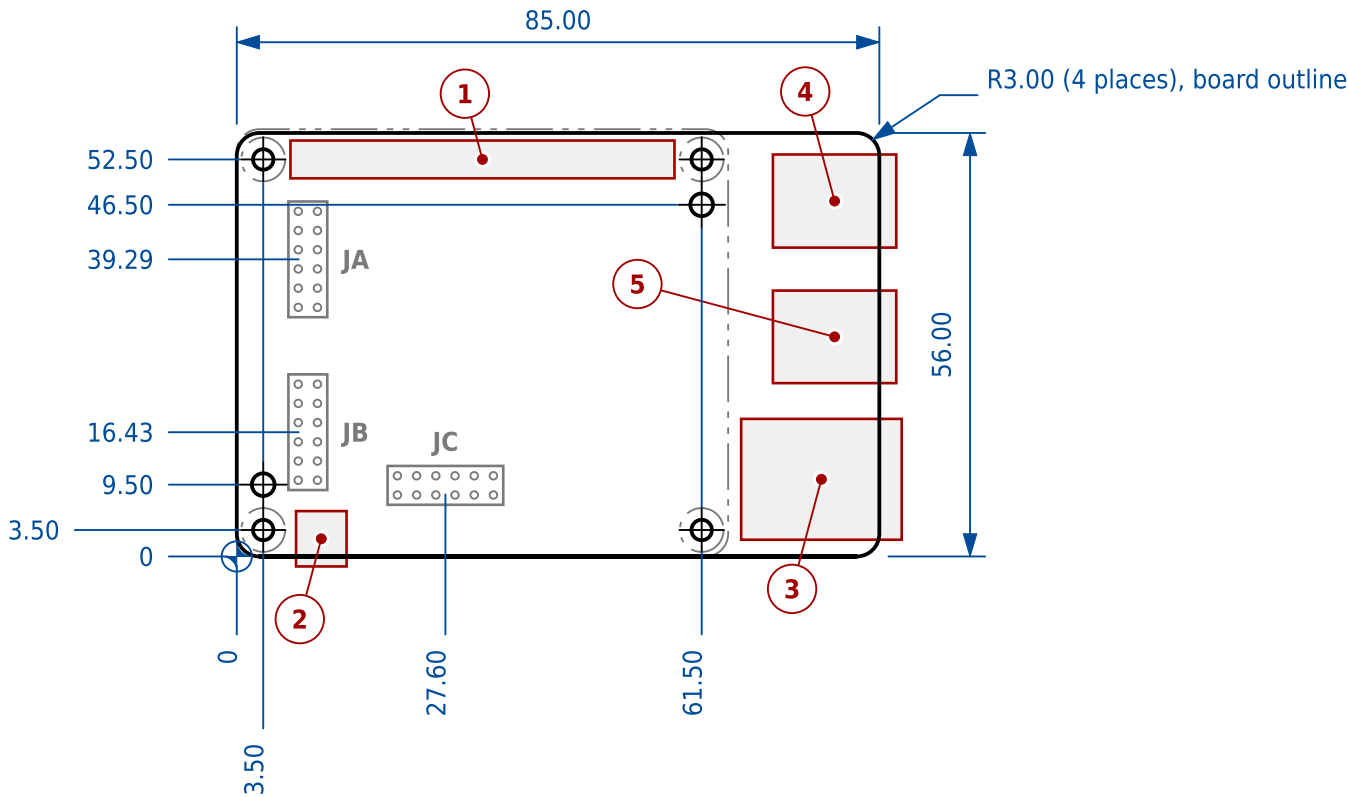
SOURCES

- Mechanical drawing: <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-mechanical-drawing.dxf> - Raspberry Pi Ltd
- Mounting holes: <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-mechanical-drawing.dxf> - Read from the DXF: hole and keep-out circles on layer 0.

RASPBERRY PI 4 MODEL B

85 x 56 mm

DRAWN generated		VERSION v0.0-579-g0e07c5a		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPI-4B				REV A
MATERIAL PCB, thickness not stated			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				



NOTES

- Viewed from the component side.
- Phantom outline is the Digilent Pmod HAT Adapter fitted on the 40-pin GPIO header; its mounting holes coincide with this board's.
- Hole IDs are assigned by this drawing and mean the same hole on every Raspberry Pi sheet.
- The 40-pin GPIO header position is fixed by the Raspberry Pi HAT specification, so it is the same on every Raspberry Pi sheet.
- Design keep-outs to 6.2 mm diameter around every mounting hole, per the Raspberry Pi HAT specification; KEEPOUT gives what this model's own drawing shows.
- Pmod host JC overhangs the lower edge, over the HDMI and power connectors. Fit the two standoffs opposite the 40-pin header, as Digilent's manual warns, so JC's pins cannot touch the HDMI shell. HDMI connectors are not drawn.
- Pmod HAT Adapter host positions are DERIVED, good to about +/-0.75 mm; drawing ACC-HAT-PMOD has the derivation.

NOTES (continued)

- Assembled envelope, connector overhang included: 87.96 x 57.32 mm.

SOURCES

- Mechanical drawing: <https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-mechanical-drawing.pdf> - Raspberry Pi Ltd
- Mounting holes: <https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-mechanical-drawing.pdf> - Hole diameter dimensioned on the drawing as $\varnothing 2.7$ mm; the keep-out circle is drawn and measures 5.80 mm.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	2.70	5.80
MT2	61.50	3.50	2.70	5.80
MT3	3.50	52.50	2.70	5.80
MT4	61.50	52.50	2.70	5.80
AUX1	3.50	9.50	3.00	not given
AUX2	61.50	46.50	3.00	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	USB-C power input	7.84 to 14.52	-1.32 to 6.00
3	Ethernet RJ45	66.72 to 87.96	2.21 to 18.19
4	USB 3.0 type A (upper pair)	70.92 to 87.24	40.84 to 53.15
5	USB 3.0 type A (lower pair)	70.92 to 87.24	22.92 to 35.16

PMOD HAT ADAPTER HOSTS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	left edge	9.400	39.290	10.670	32.940
JB	left edge	9.400	16.430	10.670	10.080
JC	bottom edge	27.600	9.400	33.950	10.670

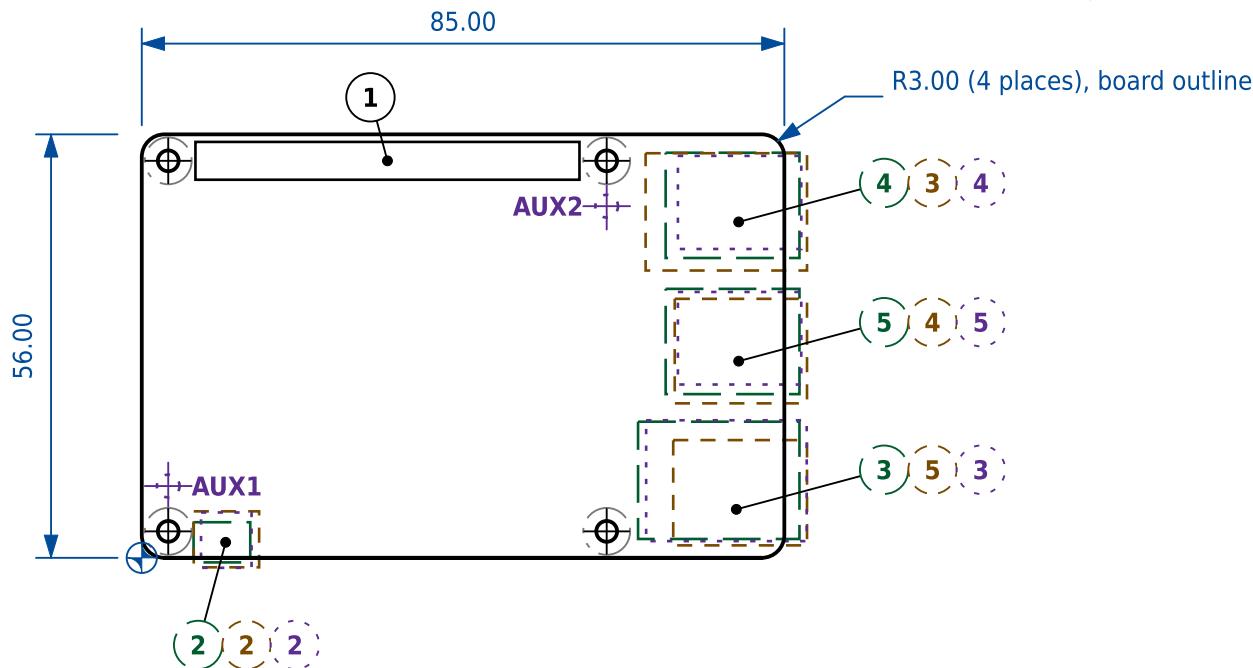
LEGEND

- Board outline
- Component body, scheduled
- Adjacent part, hole keep-out
- Dimension, extension and leader

RASPBERRY PI 5

85 x 56 mm

DRAWN generated		VERSION v0.0-579-g0e07c5a		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPI-5				REV A
MATERIAL PCB, thickness not stated			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				



NOTES

- Viewed from the component side.
- All three models share this outline exactly: 85.00 x 56.00 mm, R3.00 corners and the same four mounting hole centres. Only the hole diameter and the keep-out differ.
- The 40-pin GPIO header is in the same place on all three because the Raspberry Pi HAT specification fixes it; the three drawings were read separately and required to agree.
- The Ethernet jack and the USB type A ports swap places between the Pi 3 and the Pi 4. On the Pi 3B/3B+ and on the Pi 5 the RJ45 sits in the lower right corner with the USB ports above it; on the Pi 4B the RJ45 is in the upper right corner with the USB ports below. A chassis cut for one will not take the other.
- The Pi 5 adds AUX1 and AUX2, two 3.00 mm holes on the mounting hole axes. Power input is micro-USB on the Pi 3B/3B+ and USB-C on the Pi 4B and the Pi 5.
- The three power inputs very nearly coincide: their nearest faces are 0.025 mm apart, so at 1:1 the three outlines print as one line and that corner cannot be read as three. The feature schedule gives each model's extents.
- Assembled envelope, every model taken together and connector overhang included: 88.00 x 57.32 mm. Each model's own is on its own sheet.
- A balloon number means the same part on every Raspberry Pi sheet. Each leader goes to one place and carries a balloon for each model, left to right Pi 3B/3B+, Pi 4B, Pi 5, the ring in that model's own line type and colour, so a row reads which part each model puts there. The 40-pin header, the same part on all three, has one plain balloon.

NOTES (continued)

- Design keep-outs to 6.2 mm diameter around every mounting hole, per the Raspberry Pi HAT specification. The phantom circles are that figure; KEEPOUT gives what each model's own drawing shows.
- The Diligent Pmod HAT Adapter is not drawn. It is the same on all three, and its phantom pin fields would lie across the lower edge and the right-hand connectors, which are what this sheet exists to show; RPI-3B, RPI-4B, RPI-5 and ACC-HAT-PMOD carry it.
- HDMI, micro-HDMI and the 3.5 mm audio jack are deliberately not drawn on any sheet of this family.
- SOURCES lists the drawings only; what was read from each, and the drawing notes quoted from it, are on that model's own sheet.

SOURCES

- Pi 3B/3B+, Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.dxf>
- Pi 3B/3B+, Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-plus-mechanical-drawing.dxf>
- Pi 3B/3B+, Mounting holes: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf>
- Pi 4B, Mechanical drawing and mounting holes: <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-mechanical-drawing.dxf>

FEATURE SCHEDULE

#	ON	FEATURE	X EXTENT mm	Y EXTENT mm
1	all three	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	Pi 3B/3B+	micro-USB power input	6.85 to 14.35	-0.60 to 4.71
2	Pi 4B	USB-C power input	6.88 to 15.53	-1.25 to 6.15
2	Pi 5	USB-C power input	7.84 to 14.52	-1.32 to 6.00
3	Pi 3B/3B+	Ethernet RJ45	65.65 to 87.00	2.50 to 18.00
3	Pi 4B	Ethernet RJ45	66.65 to 88.00	37.99 to 53.51
3	Pi 5	Ethernet RJ45	66.72 to 87.96	2.21 to 18.19
4	Pi 3B/3B+	USB 2.0 type A (upper pair)	69.30 to 87.00	39.65 to 53.57
4	Pi 4B	USB 3.0 type A (upper pair)	70.50 to 88.00	20.43 to 34.25
4	Pi 5	USB 3.0 type A (upper pair)	70.92 to 87.24	40.84 to 53.15
5	Pi 3B/3B+	USB 2.0 type A (lower pair)	69.30 to 87.00	21.65 to 35.57
5	Pi 4B	USB 2.0 type A (lower pair)	70.30 to 88.00	1.65 to 15.57
5	Pi 5	USB 3.0 type A (lower pair)	70.92 to 87.24	22.92 to 35.16

HOLE SCHEDULE

ID	X mm	Y mm	ON	DIA mm 3B / 4B / 5	KEEPOUT mm 3B / 4B / 5
MT1	3.50	3.50	all three	2.75 +/-0.05 / 2.70 / 2.70	not given / 6.00 / 5.80
MT2	61.50	3.50	all three	2.75 +/-0.05 / 2.70 / 2.70	not given / 6.00 / 5.80
MT3	3.50	52.50	all three	2.75 +/-0.05 / 2.70 / 2.70	not given / 6.00 / 5.80
MT4	61.50	52.50	all three	2.75 +/-0.05 / 2.70 / 2.70	not given / 6.00 / 5.80
AUX1	3.50	9.50	Pi 5	- / - / 3.00	- / - / not given
AUX2	61.50	46.50	Pi 5	- / - / 3.00	- / - / not given

LEGEND

- Outline, mounting holes and 40-pin GPIO header: identical on all three models
- Mounting hole keep-out, 6.2 dia, per the HAT specification
- Pi 3B/3B+: power input, Ethernet and USB
- Pi 4B: power input, Ethernet and USB
- Pi 5: power, Ethernet, USB and its two extra holes
- Balloons on one leader: one per model, Pi 3B/3B+, Pi 4B, Pi 5, left to right
- Dimension, extension and callout leader

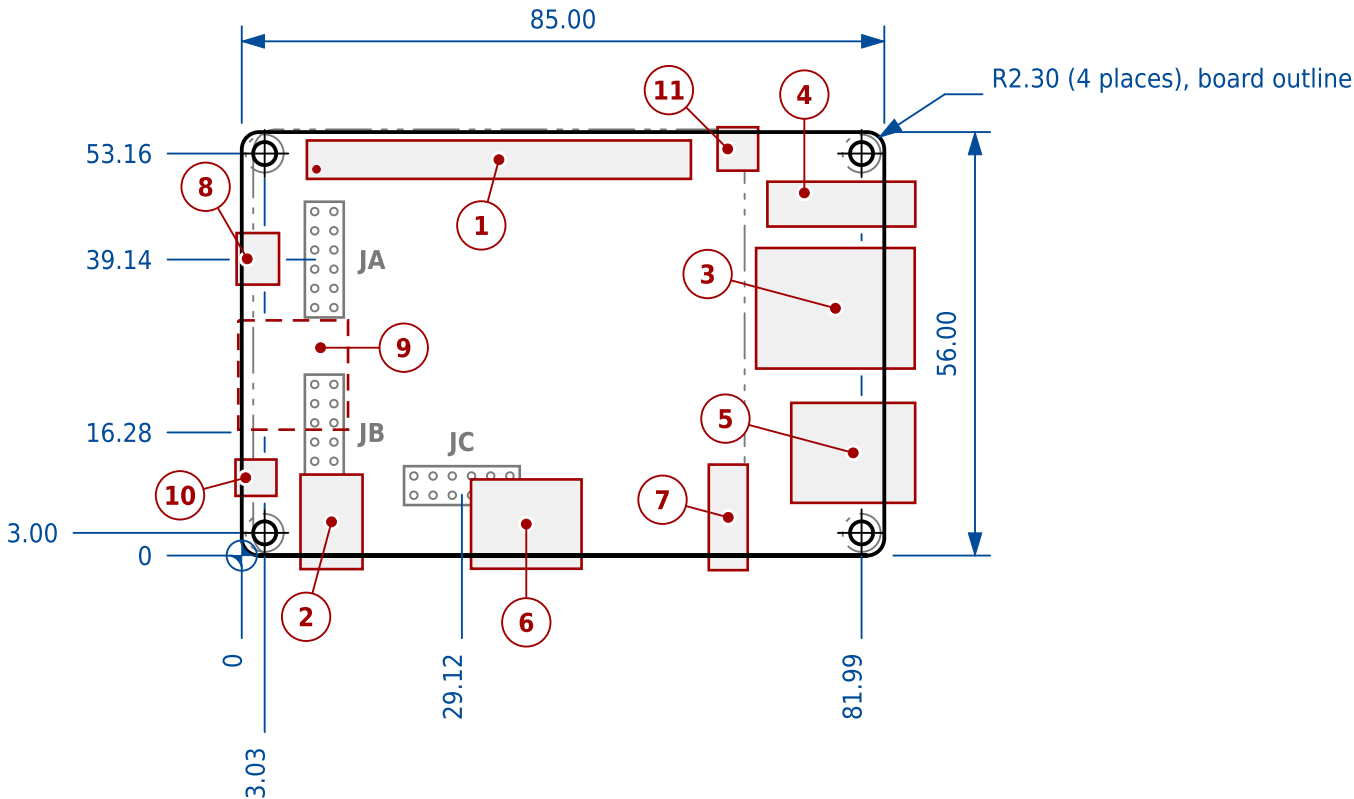
SOURCES (continued)

- Pi 5, Mechanical drawing and mounting holes: <https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-mechanical-drawing.pdf>

RASPBERRY PI 3B/3B+, 4B AND 5 COMPARED

One 85 x 56 outline: what is shared and what moves

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-369-ga4d491f	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPI-ALL	A
MATERIAL	GENERAL TOLERANCE		
- not a made part	reference only - governed by RPI-3B, RPI-4B and RPI-5		



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.03	3.00	3.05 +/-0.15	5.03
MT2	81.99	3.00	3.05 +/-0.15	5.03
MT3	3.03	53.16	3.05 +/-0.15	5.03
MT4	81.99	53.16	3.05 +/-0.15	5.03

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	40-pin GPIO header	8.62 to 59.42	49.81 to 54.89
2	DC 5 V input, barrel jack	7.77 to 15.98	-1.79 to 10.71
3	Ethernet RJ45	68.04 to 89.02	24.73 to 40.67
4	USB 2.0 type A, single, upright	69.53 to 89.10	43.51 to 49.44
5	USB 2.0 type A pair	72.69 to 89.08	6.97 to 20.19
6	HDMI	30.33 to 44.95	-1.74 to 10.07
7	3.5 mm audio and video jack	61.78 to 66.93	-1.93 to 12.03
8	Micro-USB OTG	-0.68 to 4.93	35.83 to 42.65
9	MicroSD socket, underside	-0.46 to 14.06	16.65 to 31.10
10	Power button	-0.83 to 4.59	7.89 to 12.70
11	IR receiver	62.94 to 68.31	50.90 to 56.64

PMOD HAT ADAPTER HOSTS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	left edge	10.920	39.140	12.190	32.790
JB	left edge	10.920	16.280	12.190	9.930
JC	bottom edge	29.120	9.250	35.470	10.520

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Adjacent part, hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Phantom outline is the Digilent Pmod HAT Adapter fitted on the 40-pin GPIO header; its mounting holes miss this board's by 2.0 mm or more, so no standoff can join the two.
- Xunlong publish no drawing of this board. Everything but its size is MEASURED from the photographs listed, good to +/-0.4 mm for holes and header and +/-1.1 mm for the parts: no photograph's reading, and no check on the holes and header, is further than that from what is drawn.
- Across the board the scale is the 40-pin header's 2.54 mm pitch. Fitted to the board's edges alone, the photographs came out 0.6 % wide.
- Xunlong's manual and specification table give 85 x 55 mm. The photographs measure 55.7 to 57.1 mm, and Xunlong's own dimensioned photograph and drawing of the PC Plus say 56.
- Each part is the mean of every photograph that shows it, top face from above and overhang from below; raspberry_pi/orangepi_pc.py records every reading.
- The dot is pin 1 of the 40-pin header. KEEPOUT is the copper ring round each hole.
- Plan only: no height is measured, nor how far a microSD card stands out of its socket.

NOTES (continued)

- The 40-pin header is 1.52 mm right of and 0.15 mm below a Raspberry Pi's, and the Pmod HAT Adapter moves with it. Which parts clear the adapter's underside is not established: no heights are measured.
- Pmod HAT Adapter host positions are DERIVED, good to about +/-0.75 mm; drawing ACC-HAT-PMOD has the derivation.
- Assembled envelope, connector overhang included: 89.93 x 58.57 mm.
- Corner radius measured 1.7 to 2.4 mm, drawn at the median.

SOURCES

- Board size: orangepi.org/html/hardWare/computerAndMicrocontrollers/details/Orange-Pi-PC.html - Xunlong's product page: its dimensioned photograph says 85 by 56 mm, its specification table 85 x 55. No other mechanical figure is published.
- Photographs, v1.3: linux-sunxi.org/Xunlong_Orange_Pi_PC, Orange_Pi_PC_v1.3_front.jpg and _back.jpg - Fetched from the Wayback Machine's January 2026 capture.
- Photographs, v1.2: orangepi.org/img/computersAndMmicrocontrollers/PC40.png and PC/Rectangle 641.png - Xunlong's product page.
- Measurement: raspberry_pi/measure_orangepi_pc.py - Held against other models of the board by raspberry_pi/verify_orangepi_pc.py.

XUNLONG ORANGE PI PC

Allwinner H3, v1.2 and v1.3, 85 x 56 mm

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-579-g0e07c5a	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPI-OIPPC	A
MATERIAL	GENERAL TOLERANCE		
PCB, thickness not stated	MEASURED: holes, header +/-0.4 parts +/-1.1		